

Lecture 1

Systems and Experiments

Lenny and Art wander into Hilbert's Place.

Art: *What is this, the Twilight Zone? Or some kind of fun house? I can't get my bearings*.

Lenny: *Take a breath. You'll get used to it*.

Art: *Which way is up?*

1.1 QUANTUM MECHANICS IS DIFFERENT

What is so special about quantum mechanics? Why is it so hard to understand? It would be easy to blame the "hard mathematics," and there may be some truth in that idea. But that can't be the whole story. Lots of nonphysicists are able to master classical mechanics and field theory, which also require hard mathematics.

Quantum mechanics deals with the behavior of objects so small that we humans are ill equipped to visualize them at all. Individual atoms are near the upper end of this scale in terms of size. Electrons are frequently used as objects of study. *Our sensory organs are simply not built to perceive the*